# Chapter Four

# Color images Models (RGB, HSV and YCbCr)

## 4.1 RGB Color Model

- Any color that can be represented on a computer monitor is specified by means of the three basic colors- **Red**, **Green** and **Blue** **called the RGB colors.** By mixing appropriate percentages of these basic colors, one can design almost any color one ever imagines.

  **The model of designing colors based on the intensities of their RGB components is called the RGB model, and it's a fundamental concept in computer graphics**.
- Each color, therefore, is represented by a triplet (Red, Green, and Blue), in which red, green and blue three bytes are that represent the basic color components.
- **The smallest value, 0, indicates the absence of color. The largest value, 255, indicates full intensity or saturation**.
- **The triplet (0, 0, 0) is black**, because all colors are missing, and **the triplet (255, 255, 255) is white**. Other colors have various combinations:
- For example, ( 255,0,0 ) is pure red, ( 0,255,255 ) is a pure cyan ( what one gets when green and blue are mixed ), and ( 0,128,128 ) is a mid-cyan ( a mix of mid-green and mid-blue tones ).
- **The possible combinations of the three basic color components are 256x256x256, or 16,777,216 colors**. Figure (4.1 ) shows Color specification of the RGB Color Cube.

- The process of generating colors with three basic components is based on the RGB Color cube as shown in the figure (4.1). The three dimensions of the color cube correspond to the three basic colors.

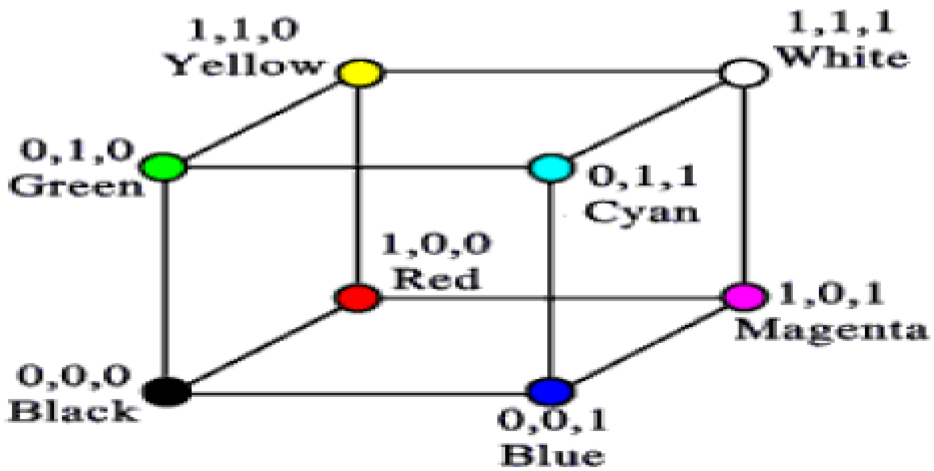


**Figure (4.1): Color specification of the RGB Color Cube**

- The cube's corners are assigned each of the three primary colors, their complements, and the colors black and white. **<u>Complementary colors are easily calculated by subtracting the Color values from 255. For example, the color (0, 0,255) is a pure blue tone</u>**.
- Its complementary color is (255-0,255-0,255-255), or (255, 255, 0), which is a pure yellow tone see Figure (4.2). Blue and Yellow are complementary colors, and they are mapped to opposite corners of the cube. The same is true for red and cyan, green and magenta, and black and white.

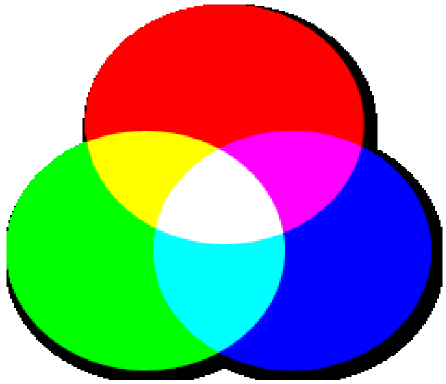

**Figure (4.2): RGB Color Model**

- Adding a color to its complement gives white.
- It is noticed that the components of the colors at the corners of the cube have either zero or full intensity.
- As we move from one corner to another along the same edge of the cube, only one of its components changes value. For example, as we move from the Green to the Yellow corner, the Red component changes from 0 to 255.
- Although we can specify more than 16 million colors, we can’t have more than 256 shades of Gray. **The reason** is that a gray tone, including the two extremes (black and white), is made up of equal values of all the three primary colors. This is seen from the RGB cube as well. Gray shades lie on the cube’s diagonal that goes from black to white. As we move along this path, all three basic components change value, but they are always equal. The value (128,128,128) is a mid-gray tone
- That's why it is wasteful ( أسراف) to store grayscale pictures using 16-million color True Color file formats.
- **Once an image is known in grayscale, we needn’t store all three bytes per pixel. One value is adequate (the other two components have the same value).**

## 4.2 HSV color Model

- HSL and HSV are two related representations of points in an RGB color space, which attempt to describe perceptual color relationships more accurately than RGB, while remaining computationally simple.
- **HSL stands for hue, saturation, lightness, while HSV stands for hue, saturation, value.**

**Hue:** the color type (such as red, blue, or yellow). a blue car reflects blue hue. The hue which is essentially the chromatic component of our perception may again be considered as weak hue or strong hue.

**Saturation:** The colorfulness of a color is described by the saturation component. For example, the color from a single monochromatic source of light ( مصدر ضوء أحادي اللون ), which produce colors of a single wavelength only , is highly saturated (مشبعا باللون للغاية ) , while the colors comprising hues of different wavelengths have little Chroma and have less saturation..

**The gray colors do not have any hue and hence they have less saturation or unsaturated. Saturation is thus a measure of colorfulness or whiteness in the color perceived.**

**The lightness (L) or intensity (I) or value (V)** : essentially provides a measure of the brightness of colors. This gives a measure of how much light is reflected from the object or how much light is emitted from a region.

The HSV model is commonly used in computer graphics applications. **The HSV color space is compatible with human color perception**

**The HSV image may be computed from RGB image using different transformation. Some of them are as follows:**

- The simplest form of HSV transformation is :

**H**= tan [3(G-B)/(R-G) +(R-B)]

**S**= 1-(min(R, G, B)/V)

**V**= (R+G+B/3)

However, the hue (H) becomes undefined when saturation S=0

- The most popular form of HSV transformation is shown next, where the r,g,b values are first obtained by normalizing each pixel such that

**r=(R/(R+G+B)), g=(G/(R+G+B)), b=(B/(R+G+B))**

- Accordingly the H, S and V value can be computed as:

**V = max (r, g, b),**

$$S = \begin{cases} 0, & if\ V = 0 \\ V - \min(r, g, b)/V, & if\ V > 0 \end{cases}$$

$$H = \begin{cases} 0, & if\ S = 0 \\ 60 * [(g - b)/S * V] & if\ V = r \\ 60\,[2 + ((b - r)/S * V)] & if\ V = g \\ 60[4 + ((r - g)/S * V)] & if\ V = b \end{cases}$$

**H = H+360 if H<0**

## 4.3 YCbCr Color Format

Another color space in which luminance and chrominance are separately represented is the $YC_bC_r$. The **Y** component takes values from 16 to 235, while Cb and Cr take values from 16 to 240. They are obtained from gamma-corrected R, G, B values as follows:

$$\begin{bmatrix} \mathbf{Y} \\ \mathbf{C_b} \\ \mathbf{C_r} \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

## 4.4 Basic Relationships between Pixels

In this section, several important relationships between pixels in a digital image are considered. As mentioned before, an image is denoted by f(x, y).When referring in this section to a particular pixel intensity.

1. **<u>Neighbors of a Pixel</u>**

**a. N4(p) : four neighbors of pixel p.**

- Any pixel p(x, y) has two vertical and two horizontal neighbors, given by: (x+1,y), (x-1, y), (x, y+1), (x, y-1), as shown in the following figure.
- This set of pixels are called the 4-neighbors of P, and is denoted by **N4(p)**
- Each of them is at a unit distance from P., and some of the neighbors of p lie outside the digital image if (x, y) is on the border of the image.

| | | |
|---|---|---|
| | f(x-1, y) | |
| f(x, y-1) | f(x, y) | f(x, y+1) |
| | f(x+1, y) | |

b. **ND(p) : four diagonal neighbors of pixel p(x,y)**, denoted by ND(p).

- have coordinates :(x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1).
- Each of them are at Euclidean distance of 1.414 from P.

c. **N8(p) These points, together with the 4-neighbors, are called the 8-neighbors of p, denoted by N8 (p).**

- As before, **some of the points in ND(p) and N8(p) fall outside the image if (x, y) is on the border of the image**.

| | | |
|---|---|---|
| F(x-1, y-1) | F(x-1, y) | F(x-1, y+1) |
| F(x, y-1) | F(x,y) | F(x, y+1) |
| F(x+1, y-1) | F(x+1, y) | F(x+1, y+1) |

$N_8$ (p)

## 2. Adjacency

- Two pixels are connected if they are neighbors and their gray levels satisfy some specified criterion of similarity.
- For example, in a binary image two pixels are connected if they are 4-neighbors and have same value (0/1)
- Let v: a set of intensity values used to define adjacency and connectivity.
- In a binary Image v={1}, if we are referring to adjacency of pixels with value 1.
- In a Gray scale image, the idea is the same, but v typically contains more elements, for example v= {180, 181, 182,....,200}.
- If the possible intensity values 0 to 255, v set could be any subset of these 256 values.